Emil R. Kee-Tui

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EDUCATION

Cornell University | Ithaca, NY
PhD in Applied Economics and Management (AEM) — May 2027

University of Arizona | Tucson, AZ
Master of Science in Applied Economics and Policy Analysis (AEPA) — May 2021

Colby College | Waterville, ME
Bachelor of Arts, Economics (major) and Mathematics (minor) — May 2019

SELECTED PROJECTS

When Pollution Rises, Reporting Falls: Coal Mining and Self-Reporting under the Safe Drinking Water Act (Job Market Paper)

Applied microeconomics, law and economics, mechanism design, natural resources, environment (abstract). Many regulations require the regulated to self-report their compliance with the law. The regulated have an incentive to reduce how much they self-report in order to hide a violation of the law or avoid the costs of self-reporting. To target enforcement, regulators may require those who are more likely to violate the law to self-report more frequently. Under this regulatory model, the most likely violators have higher self-reporting costs and greater disincentives to self-report, perversely reducing information about the most likely lawbreakers. I document this perverse outcome among U.S. drinking water utilities that are exposed to water pollution from coal mining.

Regulation Impact Assessment and Health Damage Analysis: Outdoor Worker Heat Protection Laws

Applied microeconomics, labor, health, environment, climate adaptation (abstract). Extreme heat injures outdoor workers, but firms may underinvest in protecting them from it. If they do, local heat protection rules can improve worker health, and laws that preempt them can cause harm. Texas provides a natural experiment: Austin (2010) and Dallas (2015) adopted worker heat protections, and a 2023 state law voided them and barred cities from passing new ones. We combine 38 million Texas workers' compensation claims from 2010 to 2026 with daily gridded weather data. Using difference-in-differences and event-study designs, we estimate how the adoption and repeal of the ordinances changed heat-related injuries, medical care use, and construction worker productivity.

Consumer Demand Modeling: Impact of Water Pollution on Recreation and Housing Demand

Applied microeconomics, environment, nonmarket valuation, real estate (abstract). Hedonic studies value water quality through the prices of homes on the shoreline. This misses the value lakes provide to the many people who visit them without living beside them, and so understates the benefits of cleaner water. We estimate willingness to pay for lake water quality across the contiguous U.S. in two stages. First, a recreation demand model uses cell phone visits by 46 million devices to over 2,000 lakes and satellite measures of harmful algal blooms to estimate each neighborhood's recreation value. Second, we add that value to a hedonic model of 2019–2022 home sales to recover the total value of lake water quality.

Platform Pricing and Driver Pay: Hurricanes and New York City's Ride-Hail Market

Empirical industrial organization, labor, platform markets, transportation (abstract). Storms may give ride-hail platforms more market power: when transit shuts down, riders have few alternatives, and drivers' outside options shrink. During Hurricane Ida, the driver's share of the fare fell from 85% to 73% at Uber and from 73% to 45% at Lyft. We link every New York City ride-hail trip since 2019 to rainfall and flood maps. In a stacked event study of 171 million trips in 2021, preliminary results show that fares and driver pay move together in ordinary rain, but after heavy rain fares rise faster than pay and the driver's share of the fare falls by 7 to 8 percentage points.

Competition for Space, Water, and Lithium Brine in Arkansas

Environment, natural resources, energy, water, spatial analysis (abstract). The Smackover Formation in southern Arkansas holds lithium brine estimated at 35 to 136 percent of previously documented U.S. lithium reserves. Extracting it would rely on direct lithium extraction, a technology not yet used commercially at scale, whose effects on water resources are poorly understood. We combine machine-learning estimates of dissolved lithium concentrations with watershed boundaries, EPA records of drinking water intakes, and high-resolution population data. We rank watersheds by resource potential and exposure to identify where lithium development would overlap with drinking water supplies and dense populations, giving regulators a screening tool for siting, oversight, and risk mitigation.

PROFESSIONAL EXPERIENCE

Cornell University Ithaca, New York
Graduate Resident Fellow, Carl Becker House August 2023 – Present
- Mentored, supported, and implemented extracurricular programming for 400 students in university housing.
- Trained staff in student crisis response and organized networking and extracurricular activities.
- Increased resident attendance at events by 25% and reduced student mental health calls by 10%.

Cornell University Ithaca, New York
Lead Graduate Teaching Assistant: Applied Economics and Management August 2022 – Present
- Managed five undergraduate TAs for a 200-student intermediate microeconomics class.
- Tasked with maintaining student grades throughout the semester and supporting the instructor.
- Coordinated TA office hours to focus on topics that students struggled with in assignments and exams.

World Bank Group (WBG) Washington, DC
Consultant, International Development Association June 2020 – July 2020
- Created a procedure to construct a reliable variable for crop yields from the Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA)
- Developed procedure into training document with reproducible code from data download to summary tables.
- Document adopted as a training reference.

University of Arizona Tucson, Arizona
Graduate Research Assistant for Jeffrey Michler and Anna Josephson, Assistant Professors August, 2019 – July, 2021
- Co-authored paper on market completeness in the rice market in the Philippines.
- Prepared agricultural production data from the WBG's LSMS-ISA.

Colby College Waterville, Maine
Research Assistant for Daniel Lafave, Assistant Professor June, 2017 – May, 2018
- Processed results of an RCT to reduce deforestation with improved cookstoves in rural Ethiopia.
- Co-authored an economic letter titled, 'Preferences and (Mis)Perceptions of Inequality in China.'

International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) Bulawayo, Zimbabwe
Research Assistant for Jeffrey Michler, Postdoctoral Research Fellow June, 2016 – July, 2016
- Digitized hard-copy field surveys collected for a clean cookstove adoption trial that took place in Zambia.

TECHNICAL SKILLS

Languages & Tools: Python (advanced), R (advanced), Stata (proficient), SQL (proficient), SAS (proficient), MATLAB (proficient)
Quantitative Methods: Time series econometrics, regression analysis, ML, demand modeling, hedonics, spatial modeling
Causal Inference: Difference-in-differences, instrumental variables, regression discontinuity, RCTs

TA CLASSES

AEM 2600 Managerial Economics, AEM 4190 Strategic Thinking, AEM 3205 Ethics in Business and Organizations, AEM 4421/AEM 5421 Research and Strategy in Emerging Markets, ECON 1120 Introduction to Macroeconomics

AWARDS AND PRIZES

Agricultural and Applied Economics Association *Outstanding Master's Thesis Award* May, 2022
- Won the AAEA Outstanding Master's Thesis Award for my thesis, ***Labor, Credit, and Markets: Evidence from the Philippines, 1970-2016***.
- The paper studies if small scale rice farmers in the Philippines have access to complete markets.

VOLUNTEER EXPERIENCE

Graduate Student Association: Applied Economics and Management Ithaca, New York
Co-vice president May, 2022 – June, 2023
- Assisted in designing, planning and running academic and non-academic programming for AEM graduate students.
- Organized the graduate student presentation seminar.

GRANTS

Davis Grant for independent research, Colby College Jan, 2018, and Jan 2019
- Awarded to conduct Urban Economic History Research for my senior thesis, "The Persistent Effects of Segregation on Residential Water Use in Bulawayo."
- Total Davis grant awards: $8000